🤖 Exhaustive Algorithm Setup Manual

Combining pure Al Brooks Institutional Psychology with the exact Python Mathematical Translations.

Part I: The Core Engine (`algo_engine.py`)

These algorithms run natively on the central parsing engine, detecting primary structure.

1. Opening Spikes & Early Breakouts

ALGO_ENGINE.PY

BULL SPIKE BEAR SPIKE

> **The Al Brooks Textbook Rule (Vol 2: Breakouts):** "The market is always trying to break out, then trying to make every breakout fail. A breakout is a period of CERTAINTY — both sides agree current prices are wrong; the market moves quickly to find new uncertainty. As a spike grows, risk stays constant but the target grows — this is why institutions buy relentlessly during spikes. 60%+ chance of at least a measured move."

Algorithmic Translation: Scans the first 12 bars (first hour). Drops strict relative size requirements on the first 4 bars (9:30 AM drop) to track absolute momentum, ignoring the chaos of the opening bell's gap volatility.

```
Baseline: avg_range = mean([last 20 bars range])
Threshold: 1.0x avg_range (Bars 1-4) or 1.25x avg_range (Bars 5+)
```

```
Trigger: Bar is a Strong Trend Bar AND range >= (avg_range *
threshold) or (i < 5 and bar.range >= avg_range * 0.5)
```

2. Trend Pullback Flags (H1/H2 / L1/L2)

ALGO_ENGINE.PY

H1/H2 FLAG L1/L2 FLAG

> **The Al Brooks Textbook Rule (Vol 1: Trends):** "Most pullbacks in trends have two legs (ABC corrections). High 2 in a bull trend and Low 2 in a bear trend are reliable with-trend entries. After two legs, the trend usually resumes. 80% of reversal attempts fail; most become flags. Never trade countertrend unless the trend line has been broken with momentum."

Algorithmic Translation: A rolling 5-bar counter that mathematically sums the number of pullback attempts. When the market breaks back in the trend direction, the counter issues the formal grade.

```
Pullback Definition: b.is_bear OR b.low < prev.low OR b.low <
prev2.low
Trigger: bar.high > prev.high AND the previous sequence contained
pullbacks evaluated over the last 5 bars.
```

3. Moving Average Gap Bars

ALGO_ENGINE.PY

EXHAUSTION POINT

> **The Al Brooks Textbook Rule (Vol 2: Ranges):** "20+ consecutive bars not touching the EMA = strong trend, but overdone. The first touch of the EMA is a high-

> probability scalp for a test of the trend extreme. Wait for a price action entry (reversal bar + stop) rather than just a limit order at the EMA."

Algorithmic Translation: Tracks the running total of consecutive bars remaining purely above or below the EMA. When the count exceeds 20, the engine primes itself to buy the first touch.

```
Constraint Check: Iterate last 20 bars: all(b.low > b.ema_20)
Trigger: The current bar finally touches: bar.low <= bar.ema_20
```

Part II: Custom Strategy Plugins (`user_algos/`)

Advanced institutional psychology setups programmed via the modular limits plugin.

4. Major Trend Reversal (MTR)

USER_ALGOS/BEST_SETUPS.PY

TREND CHANGE

> **The Al Brooks Textbook Rule (Vol 3: Reversals):** "The Single Most Important Rule: Never trade against a trend until there has been a breakout of the channel. A Major Trend Reversal requires: A trend, a countertrend move breaking the trend line convincingly beyond the MA, and finally, a *test* of the old extreme (a higher low or lower high) that is sharply rejected."

Algorithmic Translation: Detects the end of a dominant sequence. Requires an 80% trend adherence over 15 bars, a brutal MA violation, and a failed re-test of the absolute high/low producing a strong signal bar.

```
Prior Trend: sum(1 for b in past_15 if b.close < b.ema_20) >= 12
MA Violator: any(b.close > b.ema_20 in recent_5)
```

```
Failed Test: current.low >= trend_low * 0.995 AND it forms a
strong Bull signal bar body > range * 0.4
```

5. Breakout Pullbacks

USER_ALGOS/BEST_SETUPS.PY

BOPB TREND EXTENSION

> **The AI Brooks Textbook Rule (Vol 2: Ranges):** "All breakout pullbacks are failed breakouts that failed. These are second entries, yielding excellent mathematical probability. The 'cup and handle' is functionally a breakout pullback and is one of the most reliable with-trend setups in existence."

Algorithmic Translation: Validates that a massive breakout candle has successfully re-tested its origin wall (the former resistance which should act as new support) and survived.

```
Breakout Candle: range > base_range * 0.5 crossing the base wall.
Trigger: Current bar pulls down exactly to the wall curr.low <=
base_top AND curr.low >= (base_top * 0.998) and closes safely
above its midpoint.
```

6. Fading Weak Pullbacks (Limit Traps)

USER_ALGOS/LIMIT_SETUPS.PY

LIMIT ORDER SCALP

> **The AI Brooks Textbook Rule (Vol 1: Trends):** "Most attempts to break above or below the prior bar FAIL. Therefore, limit fading is often more reliable than stop entries. In a clear bull trend, DO NOT short L1/L2 flags. Instead, fade the weak

> shorts by placing a limit order to BUY below the prior bar's low, trapping the short sellers."

Algorithmic Translation: Identifies a steep new trend and places limit orders below counter-trend bars to actively hunt and trap weak counter-trenders taking sub-par entries.

```
Steep Trend Check: ema_now > ema_10_ago * 1.002
Weak Pullback Trigger: A bear bar whose range() < ema * 0.002 (Microscopic push).
Execution: Drops a Limit Order to BUY at curr.low - 0.01, capturing the exact price where weak bears place their sell stops.
```

7. Range Boundary Fades (2nd Entries)

USER_ALGOS/RANGE_SETUPS.PY

RANGE REVERSAL

> **The Al Brooks Textbook Rule (Vol 2: Ranges):** "80% of breakout attempts from trading ranges fail. The hallmark of a range is uncertainty. As the market approaches the bottom, bulls step aside and bears sell aggressively, creating a spike (vacuum effect). Once support is hit, both sides buy aggressively. Buy low, sell high, scalp. Only buy/short the extremes, ideally on the 2nd entry."

Algorithmic Translation: Verifies the EMA is completely flat. Waits for the absolute edge of the 20-bar box to be touched and immediately rejected.

```
Flat Box Check: abs(ema_start - ema_end) < box_height * 0.2 (EMA is horizontal).
Rejection Trigger: Price pushes to range_high * 0.998 or range_low * 1.002 but instantly closes strongly in the opposite direction.
```

8. Bear Stairs Exhaustion

USER_ALGOS/RANGE_SETUPS.PY

REVERSAL SETUP

> **The Al Brooks Textbook Rule (Vol 3: Reversals):** "Three or more pushes where each breakout is smaller shows waning momentum. Bulls take profits sooner, bears short at less extension. The 3rd or 4th push is usually the final trap before a reversal. All three-push patterns are just variants of exhausted wedges."

Algorithmic Translation: Uses calculus-style local-minima tracking in a stair-step decline. Locates the 3rd or 4th distinct lower-low that lacks follow-through.

```
Counter: Increments if a bar's low is surrounded by 2 higher lows
on both the left and right sides.
Trigger: minima >= 3 AND current bar prints a massive bull
reversal body > range * 0.5.
```